

Measuring the Hot Hand Fallacy: Do Live NFL Betting Markets Overreact to Momentum?

Assignment 5 – Second Draft

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Introduction:

Historically, sports betting in the United States has been illegal or heavily restricted. To place bets, a bettor would need to make phone calls to offshore sportsbooks in places like the Caribbean or Costa Rica. However, in 2018, the Supreme Court struck down the federal ban on sports betting, giving rise to a massive domestic market. This expansion creates a unique, real-world laboratory to test traditional economic theories against behavioral realities.

In economics, a foundational concept is the Efficient Market Hypothesis (EMH). The EMH postulates that in financial markets, the price of an asset fully reflects all available information (Fama). The consequence is that as new information is released, the price of an asset will instantly change to reflect it. This rapid adjustment process eliminates the ability to gain above-average returns in the long run (Fama).

While the EMH is generally accepted as valid for large financial markets such as the NYSE and US Treasury bonds, its application to sports betting markets is more complex. Some studies indicate that closing betting lines (the odds set right before a game begins) align with the EMH, accurately reflecting the true probability of a team winning. However, other research identifies specific situations where these markets operate inefficiently. For example, Hickman examined the NCAA Men's Basketball Tournament betting market and found that, while it operates efficiently overall, bettors exhibit biases, such as overvaluing teams from the Atlantic Coast Conference (ACC) in early rounds, leading to exploitable pricing anomalies (Hickman).

Most existing literature focuses on pre-game odds. Today, the presence of online sportsbooks has made a new market widely accessible to retail bettors: the in-game betting market. During live sports betting, new information enters the market every second as teams score and competitive dynamics change. With more information entering the market, there is a greater risk of short-term inefficiencies, as the market must continuously adjust its pricing to reflect the new state of the game. This high-frequency environment provides a novel testing ground for the EMH.

In the literature, several recognized instances of betting markets deviating from perfect efficiency are documented. This paper will focus on an inefficiency driven by the "hot hand fallacy," a cognitive bias in which individuals believe that a player who has just made a basket is more likely to make their subsequent shot. When applied to professional football, this translates into the belief that a team that completes a successful, momentum-shifting play has an artificially inflated likelihood of continued success. In their seminal research, Gilovich, Vallone, and Tversky found no statistical evidence for the hot hand hypothesis in basketball, despite strong public belief in the phenomenon (Gilovich et al.). This persistent psychological bias suggests that bettors may irrationally overvalue a team with a perceived "hot hand," causing the betting public to over-bet the "hot" team and driving the live market away from objective mathematical efficiency.

Research Question:

Do National Football League (NFL) in-game betting markets systematically overreact to momentum-shifting plays?

Data Description:

This study utilizes two primary datasets to measure pricing efficiency. The first is a high-frequency play-by-play dataset extracted using the open-source nflverse dataset via the nfl_data_py library. This provides timestamped data for every play, including game state variables such as score differential, field position, down, distance, and timeouts. This dataset includes a pre-calculated objective Win Probability metric. The second dataset consists of live in-game moneyline betting odds sourced via an odds API. While the play-by-play data is timestamped to the exact second the play occurred, the API provides snapshots of the market odds at rigid five-minute intervals.

Models and Methodology: To quantify market inefficiencies, this study measures the divergence between objective probability and subjective market pricing following a momentum event.

The Objective Win Probability (WP) Baseline

To establish the objective mathematical baseline, this study utilizes the nflfastR XGBoost WP model. The model evaluates a high-dimensional vector of in-game covariates for each play , denoted as . The raw output of the decision tree ensemble () is represented as:

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i)$$

To convert this raw output into a bounded probability between 0 and 1, the model utilizes a standard logistic sigmoid function:

$$WP_i = \frac{1}{1 + e^{-\hat{y}_i}}$$

Calculating True Implied Win Probability (IWP)

Sportsbooks charge fees to place bets and hide these profit margins in the odds that are shown. To isolate the true market sentiment, these fees, known as the overround or the vig, must be mathematically stripped from the raw odds. First, American odds are converted to a Raw IWP based on their polarity (whether the team is the favorite or the underdog):

$$\begin{aligned} \text{Raw IWP}_{\text{Favorite}} &= \frac{|\text{Odds}|}{|\text{Odds}| + 100} \\ \text{Raw IWP}_{\text{Underdog}} &= \frac{100}{\text{Odds} + 100} \end{aligned}$$

Next, the market overround is calculated by summing the Raw IWP of both teams. Finally, the True IWP is derived by dividing the team's Raw IWP by the market overround, resulting in a perfectly efficient 100 percent market total:

$$\text{True IWP}_{\text{Team}} = \frac{\text{Raw IWP}_{\text{Team}}}{\text{Overround}}$$

Measuring Momentum and Market Overreaction

This paper reflects the conceptual definition of momentum established by Roebber, Burlingame, and deWinter, who described momentum as a sustained increase in win probability by a single team over at least two consecutive changes in possession. Periods of time that match this description are where the overreaction metric will be calculated.

$$Overreaction_T = \Delta IWP_T - \Delta WP_T$$

If the change in the market's True Implied Win Probability systematically exceeds the objective change in Win Probability following intervals with high momentum generation, it will indicate a structural pricing inefficiency driven by behavioral bias.

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